MODULE 7.2

The Domains of Cloud Security

Governance Domains | Operational Domains

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Cisco Networking Academy | 2026

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Domains of Cloud Security: Governance Domains

Governance domains are strategic — the decisions an organization makes before and around adopting a cloud provider:

Domain	What It Covers
Governance & Enterprise Risk Management	Assessing the risk a cloud provider introduces to the organization's overall risk posture.
Legal Issues: Contracts & Electronic Discovery	The legal implications of storing data with a third party, often across multiple jurisdictions.
Compliance & Audit Management	Maintaining and proving regulatory compliance once the organization no longer controls the underlying infrastructure.
Information Management & Data Security	Classifying data and establishing who is responsible for protecting it once it moves to the cloud.
Interoperability & Portability	The ability to move data and workloads between providers, or back on-premises, without being locked into one vendor.

Domains of Cloud Security: Operational Domains

Operational domains are tactical — implementing and running security day to day:

Domain	What It Covers
Traditional Security, BC & DR	How moving to the cloud changes existing security operations and disaster recovery planning.
Data Center Operations	Evaluating the physical and architectural characteristics of a provider's data center.
Incident Response	How incidents are detected, communicated, and handled when the infrastructure is owned by a third party.
Application Security	Securing applications that are built for, or migrated into, a cloud environment.
Encryption & Key Management	Protecting data at rest and in transit, and deciding who controls the encryption keys.
Identity, Entitlement & Access Management	Managing identities and access policies across both the organization's own systems and the provider's.
Virtualization	Security considerations unique to running multiple VMs on shared physical infrastructure.
Security as a Service	Using third-party security services, delivered as part of the cloud, to protect cloud resources.

Abbreviations: BC = Business Continuity. DR = Disaster Recovery. VM = Virtual Machine.

Summary

- **Two categories, one framework:** the domains of cloud security split into governance domains (strategic decisions made before and around adopting a provider) and operational domains (day-to-day implementation of security controls).
- **Governance domains** revolve around risk, legal exposure, compliance, data classification, and avoiding vendor lock-in — questions an organization has to answer before signing with a cloud provider.
- **Operational domains** revolve around the practical mechanics of running security — incident response, encryption, identity management, and securing the virtualization layer itself.
- **Why this matters:** a cloud migration that only solves the operational half (good encryption, solid identity management) can still fail the organization if the governance half was never addressed — and vice versa.

References

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